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B. Tech. Degree V Semester Special Supplementary Examination August 2015

IT 1502 OPERATING SYSTEMS

(2012 Scheme)

Time: 3 Hours

Maximum Marks: 100

PART A

(Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Draw the five state process model and explain each state.
- (b) Define the term semaphore. Describe the operations associated with semaphores.
- (c) Differentiate between internal fragmentation and external fragmentation. What are the various mechanism for avoiding the fragmentation?
- (d) Draw the inverted page table architecture and describe its implementation.
- (e) Describe the functions of device drivers.
- (f) List and explain the most common file operations.
- (g) Define dead lock and write down the necessary conditions for deadlock occurrence.
- (h) Analyze the possible methods that can be adopted to handle deadlock.

PART B

(4 × 15 = 60)

- II. Compare any three process scheduling algorithms used in batch system and in interactive systems. (15)

OR

- III. (a) Design an algorithm to solve producer-consumer problem with message passing. (8)
- (b) Write down the algorithm to solve the readers and writers problem. (7)
- IV. (a) Identify the characteristic features and uses of associative memory. (7)
- (b) Compare the bitmaps and linked list memory management schemes. (8)

OR

- V. (a) Differentiate between multiprogramming with fixed partitions and variable partitions. (7)
- (b) If FIFO page replacement is used with 4 page frames and eight pages, how many page faults will occur with reference string 7103017232, if the 4 frames are initially empty? Repeat the same reference string for LRU. Also find the number of page faults. (8)
- VI. Explain the different types of disk scheduling algorithms used to handle multiple access requests. (15)

OR

- VII. Describe the operation of a DMA transfer with suitable diagram. Also explain the various operational modes of DMA. (15)
- VIII. (a) With an example describe the deadlock detection technique with one resource of each type, draw the resource allocation graph and write down the algorithms (depth first graph) used to detect the deadlock. (10)
- (b) Describe all the possible approaches that can be used to recover the deadlock situation. (5)

OR

- IX. (a) Illustrate the possible ways of preventing the occurrence of deadlock. (7)
- (b) Write Banker's algorithm for multiple resources and with suitable example prove that it can be applied to avoid deadlock. (8)